HOUSE PRICE PREDICTION USING MULTIPLE LINEAR REGRESSION

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1. Introduction and Objective

The real-estate sector needs simple and reliable tools for estimating house prices. The purpose of this project was to develop a machine learning model that predicts the price of a house in Rwanda based on its characteristics.

The dataset contained information about house area, number of bedrooms and bathrooms, house age, distance from the city centre, parking spaces, neighbourhood, and house price in million Rwandan francs (RWF).

The main objective was to build a Multiple Linear Regression model, evaluate its performance, save the trained model, and develop a Streamlit web application that allows users to enter house information and receive an estimated price.

2. Data Cleaning and Exploration

The dataset was first loaded and explored using Python, Pandas, and visualization libraries. The data types, number of rows and columns, descriptive statistics, missing values, and duplicate records were checked.

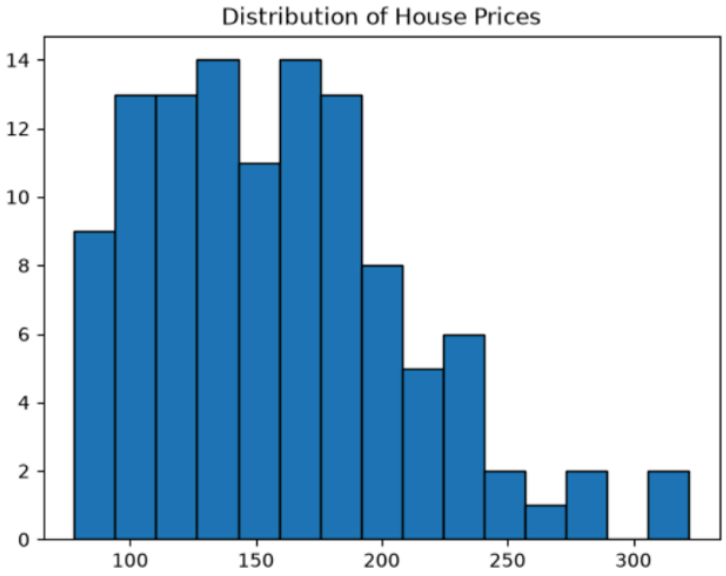
House_ID was excluded from the model because it is an identifier and does not provide useful information for predicting house prices. The numerical variables were also examined for unusual or extreme values using descriptive statistics and boxplots.

The Neighborhood column contained categorical values such as Gasabo, Huye, Kicukiro, Kigali City, Musanze, and Nyarugenge. This variable was later converted into numerical variables using one-hot encoding.

After the cleaning process, the cleaned dataset was saved as house_price_prediction_cleaned.csv.

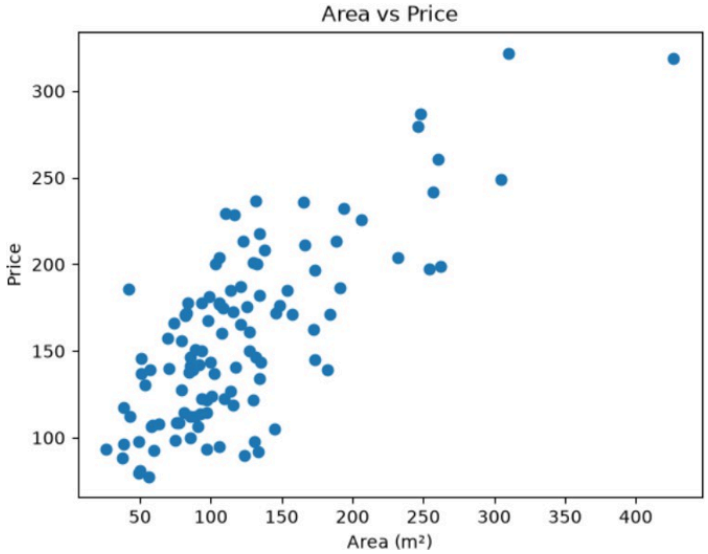
Main visualizations

Figure 1: House Price Distribution



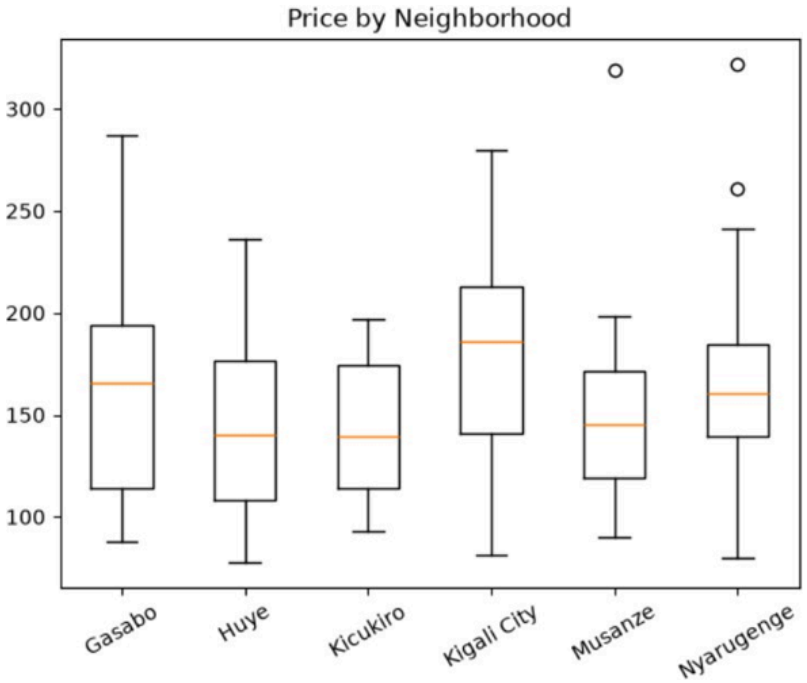
The price distribution helped show the range and spread of house prices in the dataset.

Figure 2: Area vs House Price



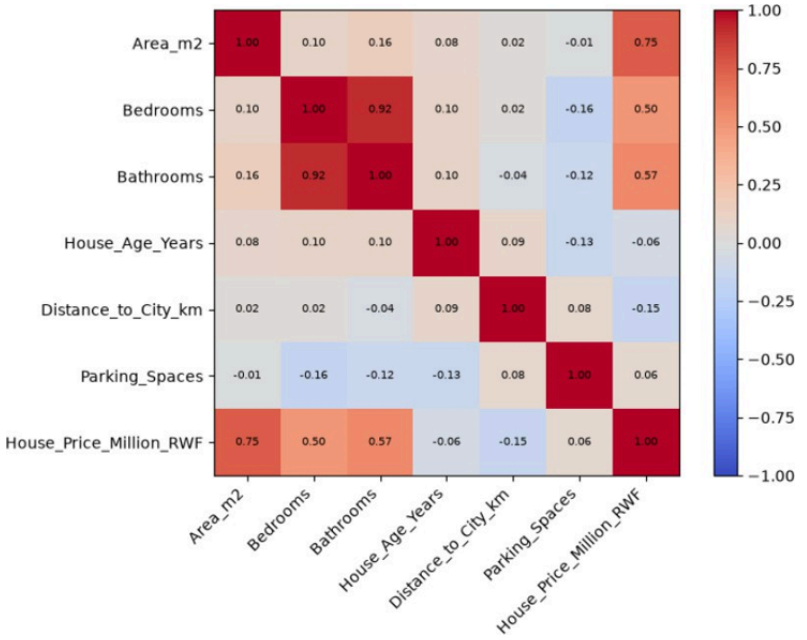
The scatter plot was used to examine the relationship between house size and price. Larger houses generally provide an important factor when estimating property value, although other variables also influence price.

Figure 3: Price by Neighborhood



The boxplot showed differences in house prices across neighbourhoods. This supported including neighbourhood as a predictor in the model.

Figure 4: Correlation Heatmap



The correlation heatmap helped identify relationships between numerical variables and house price and also provided an initial check for possible multicollinearity.

3. Model Development and Results

The target variable was:

House_Price_Million_RWF

The predictors were:

- Area_m2
- Bedrooms
- Bathrooms
- House_Age_Years
- Distance_to_City_km
- Parking_Spaces
- Neighborhood

The data was divided into 80% training data and 20% testing data, using random_state=42.

Multiple Linear Regression was selected because the target variable is continuous. A preprocessing pipeline was used to handle the categorical neighbourhood variable through one-hot encoding while keeping the model and preprocessing together.

Model Evaluation

Dataset	R2	Adjusted_R2	MAE_Million_RWF	RMSE_Million_RWF
Training	Training	0.881241	0.864493	14.493915
Testing	0.787919	0.575839	18.375411	22.936607

R^2 measures how much of the variation in house prices is explained by the model. Adjusted R^2 also considers the number of predictors. MAE represents the average absolute prediction error, while RMSE gives more weight to larger errors.

The test results provide an indication of how well the model performs on houses that were not used during training.

Coefficient Interpretation

The regression coefficients show the estimated change in predicted house price when a predictor changes while the other variables remain constant.

For example, the coefficient for Area_m2 was [insert coefficient], meaning that an increase of one square metre is associated with an estimated change of [insert coefficient] million RWF in predicted house price, holding other variables constant.

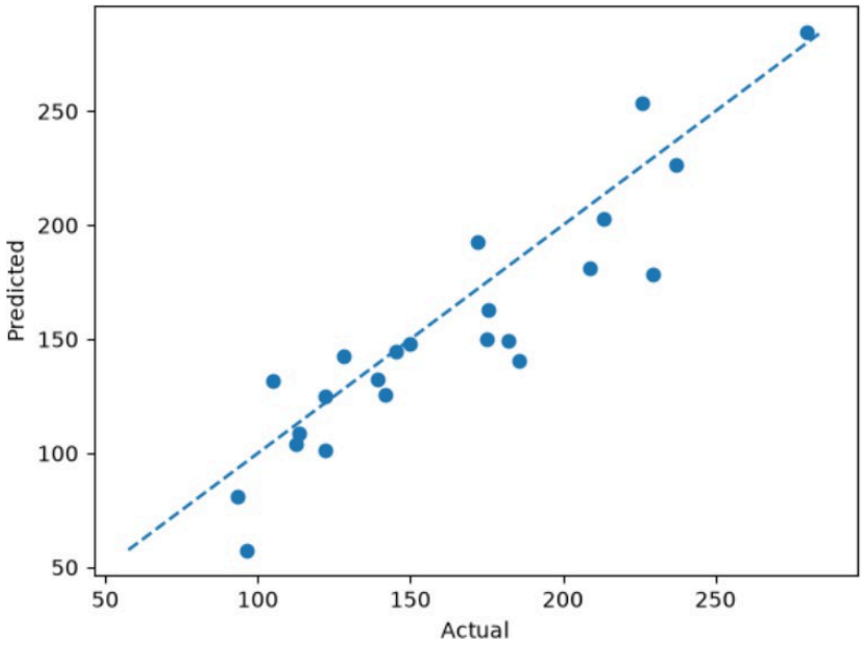
The coefficient for Bedrooms was [insert coefficient]. Therefore, an additional bedroom is associated with an estimated change of [insert coefficient] million RWF, keeping the other predictors constant.

The coefficient for Distance_to_City_km was [insert coefficient]. This indicates that increasing the distance from the city centre by one kilometre is associated with an estimated change of [insert coefficient] million RWF, holding other variables constant.

4. Model Diagnostics and Limitations

An actual-versus-predicted plot was created to compare the model's predictions with the actual house prices.

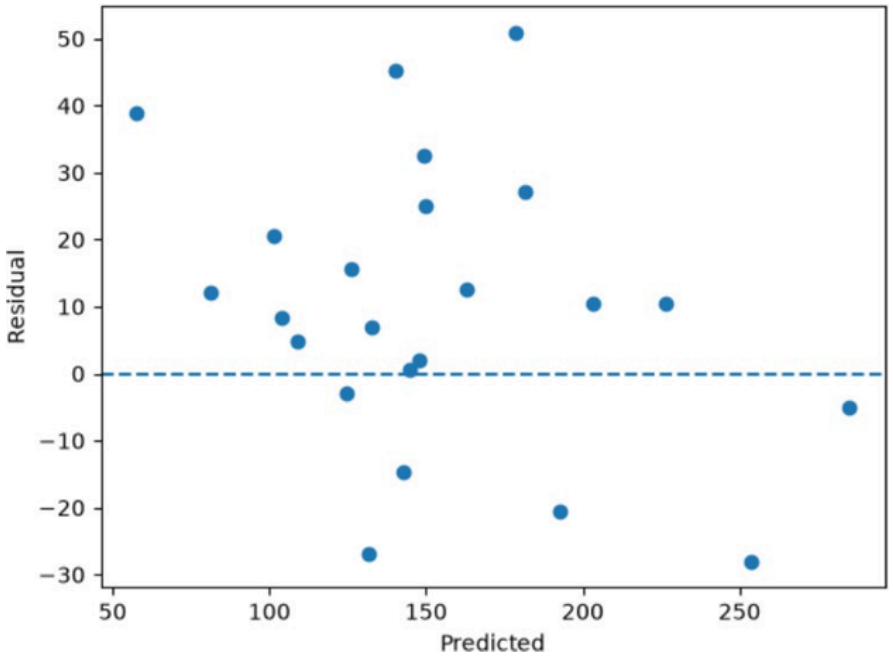
Figure 5: Actual vs Predicted House Prices



Points closer to the diagonal line indicate predictions that are closer to the actual prices.

Residual plots were also examined.

Figure 6: Residual Plot



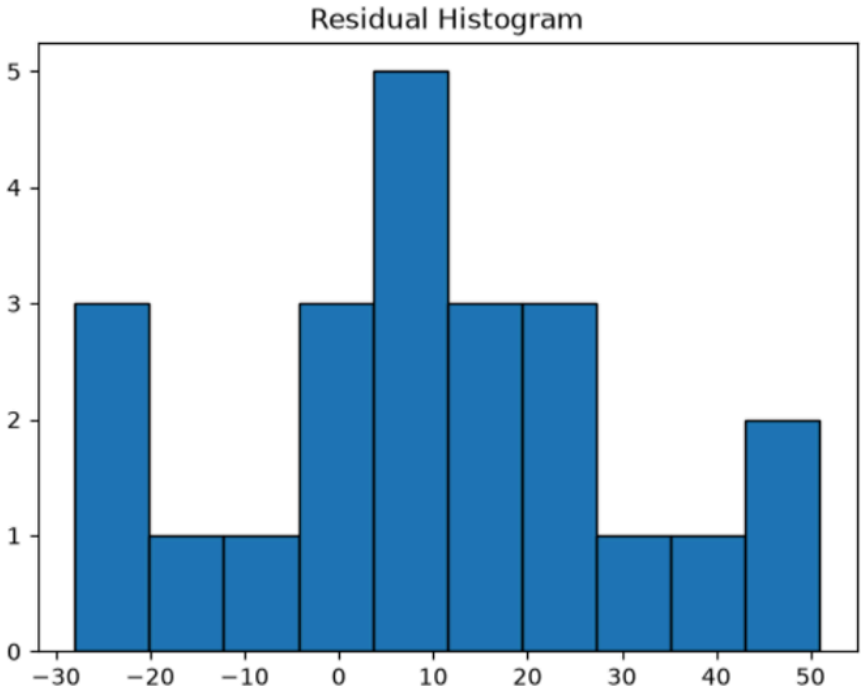


Figure 7: Residual histogram

The residual analysis was used to check whether the prediction errors showed obvious patterns. Ideally, residuals should be distributed around zero without a clear systematic pattern.

Multicollinearity was also considered using correlation analysis and/or VIF. This was important because strong relationships between predictors can make individual regression coefficients more difficult to interpret.

Limitations

The model has some limitations. First, the dataset is relatively small, so it may not represent the entire Rwandan housing market. Second, house prices can depend on other factors that were not included in the dataset, such as land size, building materials, property condition, security, access to roads, and nearby services.

Also, Multiple Linear Regression assumes that relationships between the predictors and price can be reasonably represented as linear relationships. Real-world housing prices may involve more complex relationships.

5. Streamlit Application and Deployment

The trained model was saved as:

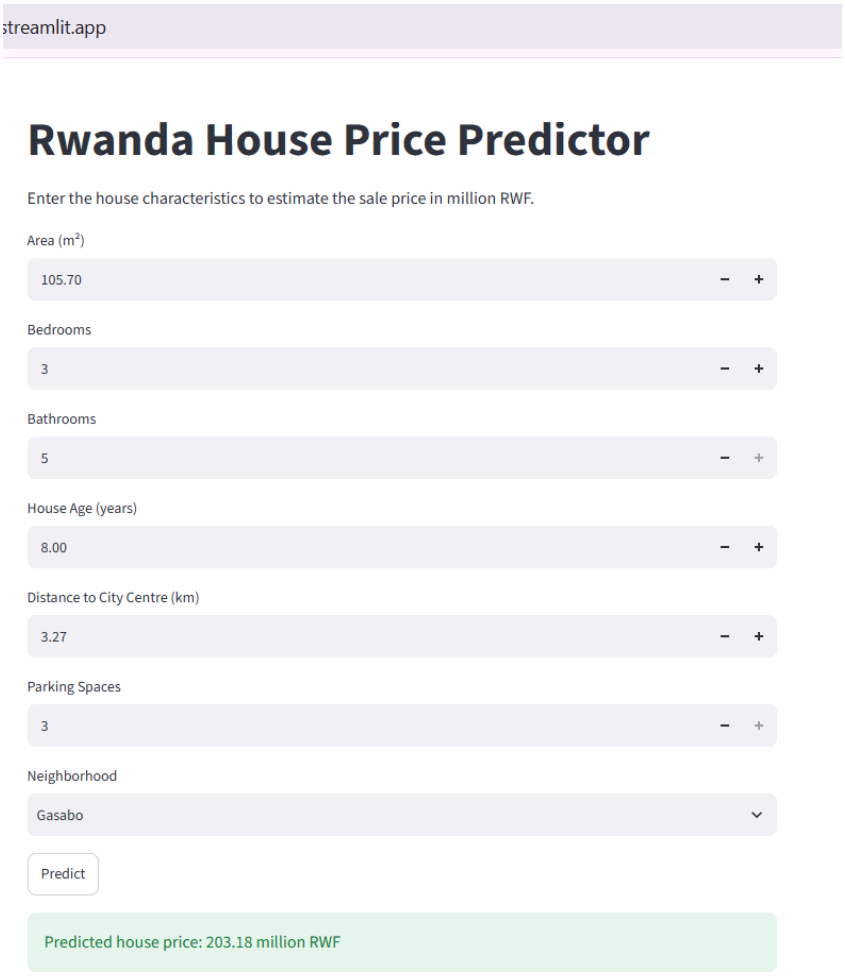
house_price_model.sav

A Streamlit application was then developed using app.py. The application allows a user to enter:

- House area
- Bedrooms
- Bathrooms
- House age
- Distance to city centre
- Parking spaces
- Neighborhood

After clicking the Predict button, the application uses the saved model to estimate the house price in million RWF.

Figure 7: Streamlit Application



GitHub Repository:
https://github.com/umwizaliza-png/house-price-predictor

Live Streamlit Application:
https://house-price-predictor-g3ckvbsrxy7hmgsuajdyh4.streamlit.app/

6. Conclusion

This project demonstrated the complete process of developing and deploying a machine learning solution for house price prediction. The dataset was explored and cleaned, relevant variables were selected, and a Multiple Linear Regression model was trained using an 80/20 train-test split.

The model was evaluated using R^2, Adjusted R^2, MAE, and RMSE, while coefficient and residual analysis were used to understand the model's behaviour. The trained model was saved as a .sav file and integrated into a Streamlit application.

The final application provides a simple way for a real-estate agent to enter house characteristics and obtain an estimated market price. Future improvements could include

collecting a larger dataset, adding more property features, and comparing the linear regression model with other machine learning algorithms.